# Online Appendix

# A A Model of Broker Advice

This appendix proves the predictions of Section 3, adds the second broker type that Section 6 estimates, and derives the consumer surplus accounting that Section 7 uses. We use the same notation as Section 3 and cite the relevant equations instead of restating them.

## A.1 The Environment

The game is the one in Section 3.1. One senior and one broker meet once. The senior enrolls in one of two plans $j \in \{L, H\}$ with annual premiums $\rho_L < \rho_H$. The game has one round: the broker recommends one plan, and the senior either enrolls in it or walks away and continues searching. Section A.7 maps the game onto the two decisions in the audit by setting parameters, and every result below holds for both.

Utility is in dollars per year. The senior pays the premium of the chosen plan and bears the fit gap $\Delta \geq 0$ only on the low-price plan, so $U_c(L) = -\rho_L - \Delta$ and $U_c(H) = -\rho_H$. The high-price plan has no fit gap by construction, so $\Delta$ measures the harm of the low-price plan relative to the high-price plan. The difference $U_c(H) - U_c(L)$ is the net harm $N$ in Equation (1).

Condition (3) describes the senior in our audit study. The coverage the low-price plan lacks is worth more to this senior than the premium it saves. As a result, the high-price plan is the plan that fits, or the plan that matches the senior's preferences, and the low-price plan is the misfit plan. We impose this condition only where stated. In Decision 2, the two plans have identical coverage, so $\Delta = 0$ (Section A.7). The net harm is then negative, and the low-price plan is the plan that fits. For this reason, we treat $\Delta$ as a free parameter rather than imposing the condition throughout.

Selling plan $j$ pays the broker $W_j > 0$ and nothing if the senior does not enroll (Section 3.1). $W_j$ is the present value (PV) of the full commission stream from the sale, the first-year payment plus the renewal payments the broker earns while the enrollee stays in the plan. The present-value convention matters for the magnitudes in Section 6.2 and not for any sign below, which requires only $W_j > 0$. Appendix E gives the term structure and the horizon we use to construct the streams.

## A.2 Perceived Fit and the Complexity Threshold

The senior perceives the premium exactly and a share $\phi \in [0, 1]$ of the fit gap (Section 3.2), so the senior acts on

$$\hat{U}_c(L) \;=\; -\rho_L - \phi\,\Delta, \qquad \hat{U}_c(H) \;=\; -\rho_H \;=\; U_c(H). \tag{28}$$

Only the low-price plan is misperceived: the senior overvalues it by $(1-\phi)\Delta$, the part of the fit gap the senior does not perceive, and values the high-price plan correctly. The difference $\hat{U}_c(H) - \hat{U}_c(L)$ is the perceived net harm $\hat{N}$ of equation (2).

**Lemma 1** (The complexity threshold)**.** *Let* $\phi^\dagger = (\rho_H - \rho_L)/\Delta$ *be the complexity threshold of equation* (9)*. Then* $\hat{U}_c(L) > \hat{U}_c(H)$ *if and only if* $\phi < \phi^\dagger$*. Under condition* (3)*,* $\phi^\dagger \in (0, 1)$*.*

*Proof.* By (28), $\hat{U}_c(L) > \hat{U}_c(H)$ if and only if $\phi\Delta < \rho_H - \rho_L$. Condition (3) gives $\Delta > \rho_H - \rho_L > 0$, so dividing by $\Delta$ gives $\phi < \phi^\dagger$, and the same two inequalities place $\phi^\dagger$ strictly between zero and one. □

When $\phi < \phi^\dagger$ the senior perceives the misfit plan to have higher utility than the plan that fits because the perceived harm $\phi\Delta$ is smaller than the premium the misfit plan saves. When $\phi \geq \phi^\dagger$ the senior correctly perceives the plan that fits to have higher utility. Lemma 1 shows that the first case is never empty under condition (3), because the threshold lies strictly below one. The threshold depends on the premium gap as well as on what the senior understands. Two seniors who perceive the same share of the fit gap can fall on opposite sides of $\phi^\dagger$, because a larger premium gap relative to the fit gap raises $\phi^\dagger$. We do not assume which case holds. Comparing the estimated $R_1$ with the calibrated $R_0$ (Section 6.2) tells us which case applies (Section A.10).

At the calibrated premium gap of $1,855 per year and the gross coverage harm of $7,429 per year, which includes the value of the option to switch to Medigap later under medical underwriting, the threshold is $\phi^\dagger = 0.25$ (Section 7.1). Using the survey's point-of-sale attribute valuations instead, $6,087 per year, which exclude that option, raises the threshold to 0.30. We use the harm with the switch option because its net value, after the premium the senior saves, is the harm that enters the broker's suitability cost in Section A.5 and the consumer surplus accounting in Section A.9. Under either harm the threshold is well below one.

We assume the senior does not infer anything about the broker's pay from the recommendation (Section 3.2). This is the most restrictive assumption in the model. In the credence-goods model of Dulleck and Kerschbamer (2006), the expert does not recommend the cheap plan because the consumer would suspect it serves the expert; our assumption rules out that inference. We impose it because the audited senior is a first-time enrollee with no prior relationship with the broker, and because Section 5.2 shows that seniors do not rate advice by its quality even after many calls. Allowing the inference would attenuate every effect derived below, and a strong enough inference would restore the canonical prediction. Note that the inference has the same sign as $\phi \geq \phi^\dagger$, so the sign of the Decision 1 response to Experiment B cannot distinguish the two. The results below hold under this assumption and do not test it.

## A.3 The Senior's Acceptance Decision

**Assumption OA.1** (The continuation value)**.** *The senior who hears a recommendation of plan $r$ either enrolls and obtains $\hat{U}_c(r)$ or walks away and obtains the continuation value $v_s$, where $s \in \{0,1\}$ indexes the perceived competition treatment (Section 3.2). The perceived competition treatment raises the continuation value, $v_1 > v_0$, and changes nothing else.*

Assumption OA.1 is the only way Experiment B enters the model. A senior who says they will solicit multiple recommendations signals a low cost of continued search, that is, a higher continuation value. Nothing on the broker's side of the model changes with $s$, and every result below uses only the inequality $v_1 > v_0$, not the size of the gap or the search process behind it.[93]

Under Assumption OA.1 the close probability of a recommendation of plan $r$ is $P_r(s) = \Lambda(\hat{U}_c(r) - v_s)$, equation (4). Since $\Lambda$ is strictly increasing, the plan the senior perceives as better is the easier sale. Lemma 1 says which plan that is, and Lemma 3 below shows that this ranking is the only feature of the senior's problem that enters the broker's decision.

**Lemma 2** (Responses of the close probabilities to the continuation value)**.** *For each plan $r$,*

$$\frac{\partial P_r}{\partial v} \;=\; -P_r\Big(1 - P_r\Big) \;<\; 0, \tag{29}$$

$$-\frac{\partial \log P_r}{\partial v} \;=\; 1 - P_r, \tag{30}$$

*and for any two plans $r$ and $r'$, $\partial \log(P_r/P_{r'})/\partial v = P_r - P_{r'}$, which for $r = H$ and $r' = L$ is equation* (10)*.*

*Proof.* Write $t_r = \hat{U}_c(r) - v_s$, so $P_r = \Lambda(t_r)$ and $\partial t_r/\partial v = -1$. Since $\Lambda' = \Lambda(1-\Lambda)$, the chain rule gives (29). Dividing by $P_r$ gives (30), and subtracting (30) for $r'$ from (30) for $r$ gives the third expression. □

A higher continuation value lowers the close probability of every recommendation, but the decline in levels, $P_r(1 - P_r)$, is largest at $P_r = \frac{1}{2}$ and need not be larger for the harder sale. The proportional decline, $1 - P_r$, is always larger for the harder sale. Equation (30) says that recommendation $r$ loses its sales at the proportional rate $1 - P_r$, the probability that the senior walks away from it, so the recommendation with the lower close probability loses sales at the faster rate. Equation (10) then says that a rise in the continuation value changes the log of the sale-probability ratio by $P_H - P_L$, so the ratio moves in favor of whichever recommendation was already an easier sale to close. The comparative statics below all follow from this derivative.

[93] The inequality follows from a stationary sequential search problem in which the senior pays a cost to contact each broker, draws a recommendation, and continues until the senior accepts one. The continuation value is then the reservation value of continued search, which falls with the contact cost, so a signal that the contact cost is low raises it.

*Remark* 1 (The logistic form is a normalization for the signs but not for the range of $R$). Under a general distribution for the taste shock, the term $1 - P_r$ in (30) becomes the reverse hazard rate of the shock distribution. The reverse hazard rate is monotone in the close probability whenever the shock density is log-concave (Bagnoli and Bergstrom, 2005), which holds for the normal, extreme-value, uniform, and Laplace distributions among others, so every sign derived below survives any of them. The scale normalization is also free, because only ratios of sale close probabilities enter the broker's decision (Lemma 3). The normalization does not extend to the range of the sale-probability ratio. The limit of $R$ as the continuation value rises depends on the tail of the shock distribution, and Remark 2 states what the logistic tail implies.

*Remark* 2 (The logistic tail places a lower bound on the sale-probability ratio). Write $a = \hat{U}_c(H)$ and $b = \hat{U}_c(L)$. Under (4), $\Lambda(a - v)/\Lambda(b - v)$ converges to $e^{a-b} = e^{\hat{N}}$ as the continuation value $v$ rises without bound. By (5), this limit is the odds ratio of the two close probabilities, $[P_H/(1 - P_H)]/[P_L/(1 - P_L)]$, which does not depend on $v$. Since $R$ is decreasing in $v$ when $P_H < P_L$ (Lemma 3), the logistic form bounds the sale-probability ratio below by that odds ratio at every continuation value. We call this lower bound the odds-ratio bound. At the survey close probabilities that calibrate $R_0$, $P_{\text{MA}} = 0.69$ and $P_{\text{MG}} = 0.42$ (Section 6.1), the odds-ratio bound is 0.33, against $R_0 = 0.61$.

We do not impose the odds-ratio bound because it is a property of the tail of the shock distribution and the audit's recommendation data do not identify that tail. Under a normal shock the same limit is zero, but under a shock with power tails it is one. The sign results of this appendix only rely on the assumption of log-concavity (Remark 1), and the odds-ratio bound is the only result that depends on the tail. The odds-ratio bound also applies to the senior's acceptance probabilities, whereas $R_1$ is the broker's belief about how likely a senior who announces plans to solicit multiple recommendations is to accept a Medigap recommendation relative to an MA recommendation (Section 3.3). The odds-ratio bound constrains that belief only if the belief follows the tail of the acceptance shock. Our secret shoppers are instructed not to accept or reject a recommendation, so the audit does not observe acceptance. Section 6.2 therefore reports $R_1$ against the bound $R_1 \geq 0$, which holds for any shock distribution because $R$ is a ratio of probabilities. The 95% likelihood-ratio interval for $R_1$ runs from zero to 0.30, which lies below the odds-ratio bound. The audit data therefore reject the odds-ratio bound at the 5% level.

## A.4 The Sale-Probability Ratio

The senior's problem enters the broker's problem through one number, the sale-probability ratio $R(s) \equiv P_H(s)/P_L(s)$ of Section 3.4. Substituting (28) into (4),

$$R(s) \;=\; \frac{\Lambda\big(-\rho_H - v_s\big)}{\Lambda\big(-\rho_L - \phi\Delta - v_s\big)}.$$

For every hundred seniors who would enroll after a recommendation of the low-price plan, $100R$ would enroll after a recommendation of the high-price plan. The perception parameter $\phi$ appears in the denominator only, because the fit gap attaches to the low-price plan alone.

**Lemma 3** (Position and derivative of the sale-probability ratio)**.** *(i) $R(s) < 1$ if and only if $\phi < \phi^{\dagger}$, and $\partial \log R/\partial\phi = \Delta(1 - P_L) > 0$.*

*(ii) $\partial \log R/\partial v = P_H - P_L$, equivalently $\partial R/\partial v = R\,(P_H - P_L)$.*

*Proof.* Since $\Lambda$ is strictly increasing, $P_H < P_L$ if and only if $\hat{U}_c(H) < \hat{U}_c(L)$, which by Lemma 1 holds if and only if $\phi < \phi^{\dagger}$, and dividing by $P_L > 0$ gives $R < 1$. For the derivative in (i), $\phi$ enters only the denominator, through the argument $t_L = -\rho_L - \phi\Delta - v_s$ with $\partial t_L/\partial\phi = -\Delta$, so by the same step as (30), $\partial \log R/\partial\phi = -\partial \log P_L/\partial\phi = \Delta(1 - P_L)$. Part (ii) is equation (10), and the level form follows from $\partial R/\partial v = R\,\partial \log R/\partial v$. □

Part (i) says that the senior's misperception reaches the broker only through the close probabilities. The unperceived share of the harm raises the probability that the senior accepts the misfit plan and leaves the probability for the suitable plan unchanged, so the broker sees the friction as the misfit plan being easier to sell. Part (ii) states what competitive pressure does. Each close probability falls at its own walk-away rate $1 - P_r$, so the sale-probability ratio declines at the rate $(1 - P_H) - (1 - P_L)$. When $\phi < \phi^{\dagger}$ the high-price plan is the harder sale, its walk-away rate is the larger one, and the sale-probability ratio falls as the continuation value rises.

*Remark* 3 (Only the ratio matters). The broker's decision, derived next, compares two expected payoffs, each a close probability multiplied by a net payment. Dividing both sides of that comparison by $P_L$ leaves the close probabilities entering only through $R$. Because the audit records recommendations and never enrollments, no close-probability level can be recovered from the audit data; however, the individual close probabilities are not needed. Since the decision is binary, the predictions, the estimation, the counterfactuals, and consumer surplus on the audited calls depend on the close probabilities only through $R$.

## A.5 The Broker's Problem

The broker's problem is the one in Section 3.3. The broker observes which plan fits the senior, makes one recommendation, is paid $W_j$ if the senior enrolls in plan $j$ and nothing otherwise, and bears the suitability cost $\kappa N$ on a completed sale of the plan the broker perceives as the misfit (equation (6)). The weight $\kappa$ the broker places on the senior's net harm is drawn on each call from a continuous distribution $F$ on the real line with density $F'$.

The suitability cost $\kappa N$ scales with the net harm the broker believes the sale does, and the broker bears it only when the senior enrolls in the plan (Section 3.3). Because the cost arrives with the sale, it is multiplied by the same close probability as the commission, so the close-probability levels drop out of the broker's rule (Proposition 1). Because $\kappa$ is unobserved, the probability of a low-price recommendation on any one call is $F(\kappa^*)$, and taking $F$ logistic turns that probability into the logit that Section 6 estimates. The logistic function in that logit describes the variation in the suitability cost across calls, not error in recording the recommendation. It also absorbs any influence on the Decision 1 recommendation that the commission streams do not capture, such as sales contests or ease of enrollment, so we allow the distribution to place mass on negative values.

For a senior whom the high-price plan fits, the low-price plan is the misfit and the broker's expected payoffs are $U_b(L) = P_L(s)(W_L - \kappa N)$ and $U_b(H) = P_H(s)W_H$. The senior's problem enters the broker's payoffs through three channels. First, the share of the harm the senior does not perceive raises only $P_L$, the probability that the senior accepts the misfit plan. Second, the continuation value lowers both $P_L$ and $P_H$, so competitive pressure changes the broker's choice only through the ratio $R = P_H/P_L$. Third, the senior's actual welfare enters only through the suitability cost, which the broker bears only on the sale of a plan that does not fit. With $\Delta = 0$ the two plans have identical coverage, neither is the misfit, and the suitability cost does not arise.

**Proposition 1** (The recommendation rule and the share of misfit recommendations)**.** *The broker recommends the low-price plan $L$ if and only if $\kappa \leq \kappa^*(s) \equiv (W_L - R(s)W_H)/N$, the cutoff of equation* (7)*, and the share of brokers who recommend the low-price plan to this senior is $\sigma(s) = F(\kappa^*(s))$, equation* (8)*.*

*Proof.* The broker recommends $L$ when $P_L(W_L - \kappa N) \geq P_H W_H$. Dividing by $P_L > 0$ gives $W_L - \kappa N \geq R(s)W_H$, and dividing by $N > 0$ gives the cutoff. Since $\kappa$ has a continuous distribution $F$, the probability that a randomly drawn broker recommends $L$ is $F(\kappa^*)$. □

Proposition 1 is Prediction 1: brokers whose weight $\kappa$ is above the cutoff recommend the plan that fits, and with no suitability cost every broker would recommend $L$ whenever $W_L > RW_H$. Note that the timing of the cost matters here. If the broker bore the suitability cost at the moment of the recommendation instead of at the sale, the comparison would be $P_L W_L - \kappa N \geq P_H W_H$, the

cutoff would be $(P_L W_L - P_H W_H)/N$, and it could not be evaluated without both close-probability levels. Because the rule is a cutoff on $\kappa$, dividing the comparison by $P_L$, by $P_H$, or by any other positive number leaves the cutoff and the share $F(\kappa^*)$ unchanged, so no normalization is needed.

Throughout the rest of this appendix, $\sigma(s)$ is the probability that a randomly drawn broker recommends the low-price plan $L$, which under condition (3) is the misfit plan, and every comparative static is a statement about that probability. Lemma 4 writes $\sigma_2$ for the share of brokers recommending the higher-priced plan in Decision 2, where $\Delta = 0$ and the low-price plan is the plan that fits.

## A.6 Predictions

Section 3.4 states three further predictions. Each differentiates the cutoff (7) with respect to one quantity the audit moves. We prove each prediction and extend Prediction 4. We then show how each prediction maps to the corresponding result in Section 5.

*Proof of Prediction 2.* Differentiating (7) with respect to $W_L$, holding $R$, $W_H$ and $N$ fixed, gives $\partial\kappa^*/\partial W_L = 1/N$, so by (8), $\partial\sigma/\partial W_L = F'(\kappa^*)/N > 0$. Differentiating with respect to the commission on the plan that fits gives the companion result $\partial\sigma/\partial W_H = -RF'(\kappa^*)/N < 0$. Neither expression contains $\phi$ or compares the close probabilities, so neither sign depends on which side of the complexity threshold the senior is on. □

Prediction 2 corresponds to Experiment A. The commission removal set the Medicare Advantage commission, which is $W_L$ in Decision 1 (Section A.7), to zero while holding the senior, the plans, and the brokers fixed, and Section 5.3 reports that the share of brokers recommending Medicare Advantage fell. Since the prediction holds on both sides of the threshold, Experiment A shows that recommendations respond to commissions. It also rules out the explanation in which every misaligned recommendation reflects limited broker knowledge, since a broker who recommends MA by mistake has no reason to change the recommendation when the pay changes. It cannot separate the explanations of Experiment B that Prediction 3 distinguishes.

*Proof of Prediction 3.* In (7) the two commissions and the net harm are fixed by the market, and Assumption OA.1 moves only the continuation value, so $R(s)$ is the only term in the cutoff that depends on $v$ and $\partial\kappa^*/\partial v = -(W_H/N)\,\partial R/\partial v$. By (8), $\partial\sigma/\partial v = F'(\kappa^*)\,\partial\kappa^*/\partial v$, and substituting $\partial R/\partial v = R(P_H - P_L)$ from Lemma 3(ii) gives equation (11). In (11), $W_H$, $R$, $N$, and $F'(\kappa^*)$ are all strictly positive, so the sign of $\partial\sigma/\partial v$ is the sign of $P_L - P_H$, which by Lemma 3(i) is the sign of $\phi^\dagger - \phi$. □

Table OA1: The response to competition by the senior's perceived harm, for a senior whom the high-price plan fits

| | Perception below the complexity threshold $P_L > P_H$, i.e. $R < 1$, i.e. $\phi < \phi^\dagger$ | Perception at or above the threshold $P_L \leq P_H$, i.e. $R \geq 1$, i.e. $\phi \geq \phi^\dagger$ |
|---|---|---|
| Commission on the misfit plan rises | share recommending it rises | share recommending it rises |
| Continuation value rises | share recommending it rises | share recommending it falls |

*Notes:* Table OA1 reports the sign of the change in the share of brokers who recommend the misfit low-price plan, for the senior whom the high-price plan fits. The first row is Prediction 2 and holds in both columns. The second row is Prediction 3 and changes sign across them. The classifying condition $P_L \gtrless P_H$ and the threshold $\phi^\dagger = (\rho_H - \rho_L)/\Delta$ are properties of one senior's problem, not of the market as a whole. Corollary 1 treats the senior whom the low-price plan fits.

When $\phi < \phi^\dagger$ the misfit plan is the easier sale. Raising the continuation value lowers the log close probability of the harder sale by more than that of the easier sale, so the sale-probability ratio falls, the cutoff rises, and the share of brokers recommending the misfit plan rises. A senior who announces an intention to compare quotes therefore raises the probability of being recommended the misfit plan. This reverses the second-opinion result of Pitchik and Schotter (1987) and Wolinsky (1993), in which a consumer who can seek another opinion makes the expert less likely to recommend the wrong product, and it is consistent with the life-insurance evidence in Anagol et al. (2017). It corresponds to the Outcome 1 result of Experiment B (Section 5.4). When $P_L = P_H$ exactly, the derivative is zero and the share does not respond to the continuation value, while the Decision 2 response of Prediction 4 remains.

**Corollary 1** (The classification is senior-specific)**.** *Consider instead a senior whom the low-price plan fits, so that $U_c(L) = -\rho_L$ and $U_c(H) = -\rho_H - \Delta'$ with $\Delta' > 0$. For that senior, $P_L > P_H$ for every $\phi \in [0, 1]$.*

*Proof.* This senior's perceived utilities are $\hat{U}_c(L) = -\rho_L$ and $\hat{U}_c(H) = -\rho_H - \phi\Delta'$. Since $\rho_L < \rho_H$ and $\phi\Delta' \geq 0$, $\hat{U}_c(L) > \hat{U}_c(H)$ for every $\phi$, and $\Lambda$ is strictly increasing. □

For this second senior the inequality at the head of the first column of Table OA1 holds for every $\phi$ and signals no friction. The cheaper plan is the better plan for that senior, so its being the easier sale is not a distortion, and because the cheaper plan also pays the broker more in Decision 1, the broker's pay and that senior's interest point the same way. A higher continuation value raises the share of brokers who recommend the low-price plan to that senior, and that plan is the plan that fits. The distinction between the two cases has content only for a senior whose fitting plan is the costlier one, which is the senior the audit constructs and the case in which the broker's pay conflicts with the senior's interest. The misinformed type in Section A.8 is a broker who has mistaken the audited senior for this second senior.

**Corollary 2** (Aligned pay does not reverse the competition response)**.** *Suppose the misfit plan's pay advantage is removed, $W_L \leq R W_H$. Then $\kappa^* \leq 0$, so the share of brokers recommending the*

*misfit plan is at most the mass of brokers with a weight at or below zero, and the magnitude of* $\partial\sigma/\partial v$ *falls with it, because* (11) *is evaluated where the density* $F'(\kappa^*)$ *is small. The sign of* $\partial\sigma/\partial v$ *is unchanged.*

*Proof.* $\kappa^* = (W_L - RW_H)/N \leq 0$ follows from $W_L \leq RW_H$ and $N > 0$. The statements about $\sigma$ and its derivative follow from (8) and (11), whose sign does not depend on where the density is evaluated. □

Corollary 2 separates the commission from the competition response. Cutting the misfit plan's commission lowers both the share of brokers recommending it and the sensitivity of that share to the continuation value, because it moves brokers away from the cutoff at which the continuation value acts. The sign of the competition response depends only on whether the senior's perception is above or below $\phi^\dagger$; no commission schedule can change it. The audit contains the aligned-pay case directly, since the commission removal of Experiment A sets $W_L = 0$, and Proposition 3 gives the model's prediction there.

**Lemma 4** (Decision 2: the recommended markup)**.** *Let* $\Delta = 0$ *and* $W_j = z\rho_j$ *for a commission rate* $z > 0$*. Then, for every* $\phi \in [0, 1]$*:*

*(i) the broker recommends the higher-priced plan if and only if* $R(s) \geq \rho_L/\rho_H$*;*

*(ii)* $P_H < P_L$*, so the cheaper plan is the easier sale;*

*(iii) a broker who recommends the cheaper plan at continuation value* $v$ *recommends the cheaper plan at every* $v' > v$*;*

*(iv) the share of brokers recommending the higher-priced plan is non-increasing in the continuation value, and strictly decreasing whenever a positive density of choice sets is at the switching threshold, and the mean markup of the recommended plans falls with it.*

*Parts (i) and (iv) are Prediction 4.*

*Proof.* (i) With $\Delta = 0$ neither plan is the misfit, no sale carries the suitability cost, and the payoffs are $U_b(L) = P_L z\rho_L$ and $U_b(H) = P_H z\rho_H$. The broker recommends the higher-priced plan when $P_H z\rho_H \geq P_L z\rho_L$, and dividing by $zP_L\rho_H > 0$ gives $R(s) \geq \rho_L/\rho_H$. (ii) With $\Delta = 0$, (28) gives $\hat{U}_c(L) = -\rho_L > -\rho_H = \hat{U}_c(H)$ for every $\phi$, so $P_H < P_L$. (iii) Let $D(v) = \log R(v) - \log(\rho_L/\rho_H)$, so that by (i) the broker recommends the higher-priced plan if and only if $D(v) \geq 0$. By Lemma 3(ii) and part (ii), $\partial D/\partial v = P_H - P_L < 0$, so once $D$ falls below zero it stays below zero. (iv) Across choice sets, the share of brokers recommending the higher-priced plan is $\sigma_2(v) = \Pr\{D(v) \geq 0\}$. By (iii) the set of choice sets on which $D(v) \geq 0$ shrinks as $v$ rises, so $\sigma_2$ is non-increasing, and strictly decreasing whenever the distribution of $D(v)$ across choice sets has positive density at zero.

The markup of a recommendation is $\rho_H - \rho_L$ when the higher-priced plan is recommended and zero otherwise, so the mean recommended markup falls with $\sigma_2$. □

Prediction 4 corresponds to Outcome 2 of Experiment B. Section 5.4 reports that the treatment lowered the mean annual markup of the Medigap plans brokers recommended. The result holds for every value of $\phi$, so it serves as the control experiment for Prediction 3: the same increase in the continuation value, applied to a decision in which nothing is hidden, lowers the recommended premium on either side of the complexity threshold. The Decision 2 response also exists only because Medigap commissions are proportional to the premium. With a flat commission, part (i) would read $R \geq 1$, which by part (ii) never holds, and no marked-up plan would ever be recommended. The framework has two plans per choice set; the estimated model in Section 6 uses the senior's full choice set, and the intuition behind part (iv) carries over.

*Remark* 4 (The joint pattern rules out commission maximization). Predictions 3 and 4 describe one movement in both decisions: under competitive pressure, the broker moves toward the plan the senior is more likely to accept. The estimate $\hat{R}_1 < R_0$ implies that the senior's friction parameter $\phi$ is below $\phi^\dagger$, so this is the cheaper plan in both decisions. Because the two decisions order commissions in opposite directions (Table 2), this movement raises the broker's expected commission in Decision 1 and lowers it in Decision 2.

The pair of results rules out a simple alternative: that the announcement leads the broker to push whichever plan pays the larger commission. Such a broker would move toward Medicare Advantage in Decision 1, as we observe, but toward the higher-priced Medigap plan in Decision 2, the opposite of what we observe. Decision 1 alone cannot distinguish the two explanations, because both predict a move toward Medicare Advantage. Decision 2 can because the two explanations predict opposite moves in Decision 2.

One alternative is consistent with both results. The broker may read the announcement as a sign that the caller cares mostly about price, and recommend the cheaper plan in both decisions for that reason. This explanation predicts the same directions as ours, but through a change in what the broker believes the caller prefers rather than in how likely the caller is to walk away. Section 5.4 tests it directly and finds that the announcement does not make the caller appear more price-sensitive.

## A.7 The Two Audit Study Decisions

The two decisions in the audit are the game of Sections A.1–A.5 under the two parameterizations in Table 2. In Decision 1 the broker chooses between recommending Medicare Advantage and recommending Traditional Medicare plus a Medigap plan. Setting $\Delta = \Delta^{\mathrm{MA}} > 0$, the gross under-insurance harm of Medicare Advantage for the audited senior, and $W_L > W_H$, everything

in Sections A.5 and A.6 applies without modification. In Decision 2 the broker chooses among coverage-identical Medigap plans in the senior's choice set. Setting $\Delta = 0$ and $W_j = z\rho_j$ gives Prediction 4. Two further consequences follow. First, the suitability cost does not arise, because the plans are coverage-identical and neither is the misfit, so in the framework Decision 2 carries no cost of recommending the wrong plan and in the estimated model it does not depend on the broker's type. Second, the broker's rule contains no parameters once $z$ cancels (Lemma 4(i)). The estimated model does give Decision 2 a cost of its own, the Decision 2 suitability cost $\kappa_2$ in equation (18), which penalizes the premium the senior pays above the cheapest coverage-identical plan and not a loss of coverage. Appendix E explains why the two suitability costs are distinct parameters.

## A.8 Broker Types

The conceptual framework described above assumes the broker knows which plan matches the senior's preferences. Experiment A shows that this is not always true. The estimated model in Section 6 accounts for this with a latent broker type.

**Assumption OA.2** (Broker types)**.** *Each broker $b$ has a fixed type $\tau_b \in \{I, M\}$. The informed type $I$ knows that the high-price plan fits the audited senior. The misinformed type $M$ believes that the low-price plan fits the audited senior, so the plan that type perceives as the misfit is the high-price plan. The informed type perceives the net harm $N^{MA} = \Delta^{MA} - (\rho_H - \rho_L)$ from an MA sale: the gross under-insurance harm less the premium the senior saves on MA, as in* (1)*. The misinformed type perceives the net harm $N^{MG} = \Delta' + (\rho_H - \rho_L)$ from a Medigap sale: the over-insurance harm that type perceives plus the premium the senior pays above MA. The probability of being misinformed, $\lambda(x_b)$, is a logistic function of broker characteristics fixed before the call.*

The two types believe they are doing different damage, so the assumption lets each type's suitability cost scale with its own perceived harm. A broker of the informed type who recommends Medicare Advantage to the audited senior perceives the net harm $N^{\mathrm{MA}} = \$5{,}574$ per year. This is the survey's gross coverage harm of \$7,429 per year, which includes the forgone option to switch to Medigap later under medical underwriting, net of the \$1,855 premium the senior saves; it is the net harm the consumer surplus accounting uses. A broker of the misinformed type recommends MA on most calls whether or not MA pays a commission (83% of no-MA-commission calls at the estimates), although that broker's recommendation still responds to both commissions (equation (17)). The harm cost this type perceives from a Medigap sale may be better described as a perceived cost of effort or an information cost of recommending a Medigap plan. For the senior in Corollary 1, where the low-price plan fits, this net harm is $N^{\mathrm{MG}} = \Delta' + (\rho_H - \rho_L)$. The main specification sets $N^{\mathrm{MG}} = N^{\mathrm{MA}}$ (Appendix E).

**Proposition 2** (The cutoff rules of the two broker types)**.** *Under Assumption OA.2, a broker of*

*the informed type recommends the low-price plan if and only if* $\kappa \leq \kappa_I^* \equiv (W_L - R(s)W_H)/N^{MA}$, *the cutoff* (7) *with* $N = N^{MA}$, *and a broker of the misinformed type recommends the low-price plan if and only if*

$$\kappa \;\geq\; \kappa_M^* \;\equiv\; \frac{W_H - W_L/R(s)}{N^{MG}}. \tag{31}$$

*Proof.* For the informed type the perceived misfit is the low-price plan, and Proposition 1 applies with $N = N^{\mathrm{MA}}$. For the misinformed type the perceived misfit is the high-price plan, so $U_b(L) = P_L W_L$ and $U_b(H) = P_H(W_H - \kappa N^{\mathrm{MG}})$. The misinformed type recommends the low-price plan when $P_L W_L \geq P_H(W_H - \kappa N^{\mathrm{MG}})$. Dividing by $P_H > 0$, using $P_L/P_H = 1/R(s)$, and rearranging gives $\kappa N^{\mathrm{MG}} \geq W_H - W_L/R(s)$, and dividing by $N^{\mathrm{MG}} > 0$ gives (31). □

In the dollar units of Section 6.1, the informed type's rule is equation (16), and the two rules are the two lines of equation (17). The two rules differ in two respects. First, the inequality reverses, because a higher suitability cost pushes the informed type away from the low-price plan and pushes the misinformed type toward it. Second, the sale-probability ratio multiplies $W_H$ in the informed type's rule and enters as $1/R$ in the misinformed type's rule, because each type divides the payoff comparison by the close probability of the plan it perceives as the misfit.

**Proposition 3** (The no-MA-commission floor)**.** *Set* $W_L = 0$, *the commission-removal case of Experiment A. Then:*

*(i) the informed type's cutoff is* $\kappa_I^* = -R(s)W_H/N^{MA} < 0$, *so the share of brokers of the informed type recommending the low-price plan is* $F(\kappa_I^*)$, *the mass of brokers with a weight below a strictly negative cutoff, which is small when* $F$ *places little mass below zero;*

*(ii) the misinformed type's cutoff is* $\kappa_M^* = W_H/N^{MG}$, *which does not contain the sale-probability ratio;*

*(iii) the perceived competition treatment therefore has no effect on the share of brokers of the misinformed type recommending the low-price plan, and the observed share of brokers recommending the low-price plan at zero commission is* $\bar{\lambda}\big(1 - F(\kappa_M^*)\big) + (1 - \bar{\lambda})F(\kappa_I^*)$.

*Proof.* Setting $W_L = 0$ in $\kappa_I^*$ and in (31) gives (i) and (ii), since $R$, $W_H$, $N^{\mathrm{MA}}$ and $N^{\mathrm{MG}}$ are strictly positive. The sale-probability ratio enters (31) only through $W_L/R(s)$, which is zero when $W_L = 0$: the close probability of the Medigap sale multiplies both the commission from that sale and the cost the misinformed type attaches to it, so it cancels. For (iii), the misinformed cutoff does not depend on $s$, and by Assumption OA.1 the treatment moves nothing else in the misinformed type's problem. With probability $\bar{\lambda}$ the broker is of the misinformed type and recommends the low-price plan when $\kappa \geq \kappa_M^*$, which has probability $1 - F(\kappa_M^*)$, and with probability $1 - \bar{\lambda}$ the broker is of the informed type and recommends it with probability $F(\kappa_I^*)$. □

Proposition 3 identifies the misinformed share from the difference between the two types at a zero MA commission. At $W_L = 0$ a broker of the informed type who recommends the low-price plan gives up the commission on the plan that fits and bears the suitability cost on the plan that does not, so the informed type's probability of recommending the low-price plan is close to zero. A broker of the misinformed type faces a different trade-off. That broker believes Medigap is the misfit plan, so a Medigap recommendation earns the Medigap commission but carries the suitability cost, and an MA recommendation earns nothing but carries no cost. The misinformed type therefore recommends MA whenever the suitability cost exceeds the Medigap commission, $\kappa N^{\mathrm{MG}} \geq W_H$, and the misinformed type's cutoff is positive. The observed rate at which brokers recommend MA when it pays nothing is approximately the misinformed share multiplied by the misinformed type's MA recommendation rate, which is high but below one.

At the estimated parameters, brokers of the misinformed type recommend MA on 83% of no-MA-commission calls and brokers of the informed type on 2%. The observed rate of 23% is a lower bound on the misinformed share if and only if the misinformed type's Medigap recommendations outnumber the informed type's MA recommendations on no-commission calls. The condition holds at the estimates. The misinformed type recommends Medigap on 17% of these calls, and the informed type recommends MA on only 2%. If the misinformed share were small enough that the informed type's MA recommendations dominated, the observed rate would overstate the share. Section 5.3.2 states the bias in each direction. Section 6.2 recovers $\bar{\lambda}$ from this floor through the distribution of the suitability cost.

## A.9 Consumer Surplus

The predictions sign the broker's responses. Converting them into consumer surplus for the senior requires one further assumption.

**Assumption OA.3** (Recommendation compliance)**.** *For the consumer surplus accounting, the senior enrolls in the plan the broker recommends.*

This is a reasonable approximation in this market because the broker enrolls the beneficiary during the call (Section 4), so the recommendation and the enrollment are the same event far more often than in advice markets where the consumer transacts elsewhere. It is also the reason the close-probability levels, which Remark 3 shows are neither identified nor needed for the predictions, are not needed here either.

Write $\sigma$ for the share of brokers recommending Medicare Advantage in Decision 1, $\sigma_2$ for the share recommending the higher-priced Medigap plan in Decision 2, $\rho_c$ for the premium of the cheapest coverage-identical Medigap plan in the senior's choice set, and $m$ for that choice set's markup. Under Assumption OA.3 the audited senior's expected utility is $\sigma\, U_c(L) + (1-\sigma)(-\rho_c - m\sigma_2)$. With probability $\sigma$ the senior is recommended Medicare Advantage and receives the utility

of the misfit plan. With probability $1-\sigma$ the senior is recommended Medigap, pays the cheapest coverage-identical premium, and pays the markup on the share $\sigma_2$ of those calls on which the broker recommends a higher-priced plan. Measured against the benchmark in which the senior receives the cheapest plan that fits, which gives $-\rho_c$, the expected loss is

$$\mathcal{L} \;=\; \sigma\, N \;+\; (1-\sigma)\, m\, \sigma_2, \tag{32}$$

where $N$ is the net harm of (1) evaluated at $\rho_H = \rho_c$. The first term is the misallocation loss: the share of calls on which the senior is steered into the wrong product, multiplied by the dollars that error costs the senior per year. The second is the overpayment loss: the share of calls on which the senior is recommended the right product but a needlessly expensive version of it, multiplied by the annual overpayment. Section 7 evaluates (32) at the estimated parameters, per Medigap-fit senior per year.

*Remark* 5 (The model does not sign the consumer surplus effect of perceived competition). Differentiating (32) with respect to the continuation value,

$$\frac{\partial \mathcal{L}}{\partial v} \;=\; \frac{\partial \sigma}{\partial v}\big(N - m\,\sigma_2\big) \;+\; (1-\sigma)\, m\, \frac{\partial \sigma_2}{\partial v}. \tag{33}$$

When $\phi < \phi^{\dagger}$, the first term is positive. Prediction 3 gives $\partial\sigma/\partial v > 0$, and the net harm exceeds the mean markup at the calibrated values. The second term is negative by Prediction 4. Which term is larger depends on the estimated parameters. For this reason, we report the consumer surplus effect of Experiment B as a counterfactual in Section 7 rather than as a prediction.

*Remark* 6 (Transfers and misallocation). The two terms of (32) enter the two surplus measures differently. The markup is a transfer from the senior to the broker and the insurance company, so it enters the consumer surplus loss in full and cancels from total surplus. Commissions are transfers from insurance companies to brokers and also cancel from total surplus. The misallocation term $\sigma N$ enters both. A Medigap-fit senior steered into Medicare Advantage loses coverage value and the option to switch to Medigap later, net of the premium gap the senior saves, and no other party gains that amount. A policy that acts only on Decision 2 therefore raises consumer surplus and not total surplus, while a policy that acts only on Decision 1 raises both.

## A.10 Mapping to the Structural Model

The objects in this appendix enter the estimated model in Section 6 as follows. The weight $\kappa$ and its distribution $F$ become the Decision 1 suitability cost $K_{1,i} = \kappa_{1,i}\mathcal{A}N$ (the net harm enters in present value, $\mathcal{A}N$), which is logistic with location $\bar{K}_{1,i}$ and scale $\mu_K = \$1{,}000/\beta$, both in PV dollars, and Table 9 lists the estimated and calibrated parameters. The net harm $N$ does not enter the likelihood. It values each averted MA recommendation in the consumer surplus accounting

(Remark 6). We note two features of the mapping. First, no parameter of the senior's problem is estimated. The perception parameter $\phi$, the continuation value $v_s$, and the close probabilities enter the broker's decision only through the sale-probability ratio, and the estimation works directly with that ratio. Second, the direction of the competition response is not imposed. We calibrate the baseline sale-probability ratio $R_0$ from the broker survey and estimate the treated ratio $R_1$ against it, so the question of which side of the complexity threshold the senior is on, the two columns of Table OA1, becomes the testable statement $\hat{R}_1 < R_0$.

## A.11 Model limitations and simplifying assumptions

For tractability, we make three simplifying assumptions. First, we assume a broker makes a single recommendation to a single senior and the game ends, meaning reputation and repeat business enter only through the weight $\kappa$. Second, we assume insurance companies set premiums and commissions outside the model. Thus, the counterfactuals estimated in Section 7 are partial equilibrium. Finally, we assume seniors do not learn about policy changes such as commission changes or update their beliefs in response to them. Section 5.2 suggests this is a better approximation in this market than a fully informed senior.